



**S. B. JAIN INSTITUTE OF TECHNOLOGY,
MANAGEMENT & RESEARCH, NAGPUR.**

Practical No. 12 (Post Lab)

Aim: To explore the features and capabilities of Luma AI for generating and editing images using artificial intelligence.

Name of Student:

Roll No.:

Semester/Year: V/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: To explore the features and capabilities of Luma AI for generating and editing images using artificial intelligence.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

- To understand the basic concept of AI-based image generation.
- To learn how to write effective prompts for generating desired visuals.
- To explore different features and capabilities available in Luma AI.
- To distinguish between simple and well-structured prompts.
- To generate and refine images or 3D visuals using AI tools.
- To understand the role of AI in creativity, design, and digital content creation.

Theory:

Artificial Intelligence (AI) has revolutionized digital content creation by enabling machines to generate images, videos, and 3D models based on user inputs. One such advanced tool is Luma AI, which focuses on AI-powered 3D content generation, photorealistic rendering, and scene reconstruction.

Luma AI uses advanced deep learning techniques such as **NeRF (Neural Radiance Fields)** to capture real-world objects and environments and convert them into highly detailed 3D models. It allows users to create realistic scenes from simple inputs like photos or videos. The platform also supports text-based generation, where prompts can be used to generate visuals.

The concept of **prompt-based generation** plays a role in guiding AI outputs. Users describe what they want, and the AI interprets the input using natural language processing to generate results. A well-structured prompt with details about lighting, perspective, textures, and style leads to better results.

Another important concept is **prompt engineering**, which involves crafting precise and descriptive inputs. By specifying attributes like environment, camera angle, realism, and lighting conditions, users can control the final output more effectively.

Luma AI provides features such as:

- 3D scene generation
- Image-to-3D conversion
- Realistic rendering
- Scene editing and refinement

This technology is widely used in industries like gaming, architecture, film production, virtual reality, and digital marketing. It reduces manual effort while enhancing creativity and productivity.

In conclusion, Luma AI helps users understand AI-driven visual generation, improves prompt design skills, and provides hands-on experience in modern AI-based creative workflows.

USES OF LUMA AI:

- Generates realistic 3D models from images or videos
- Creates immersive environments and scenes
- Useful for AR/VR content development
- Helps in product visualization and prototyping
- Used in filmmaking and animation
- Assists in architectural and interior design

HOW TO USE LUMA AI:

- **Step 1:** Create an account and log in
- **Step 2:** Capture images/videos or enter a prompt
- **Step 3:** Upload data or select generation mode
- **Step 4:** Process the input using AI
- **Step 5:** View generated 3D model or image
- **Step 6:** Edit or refine using available tools

APPLICATIONS OF LUMA AI:

- **Game Development** – 3D assets and environments
- **Architecture** – building visualization
- **Film & Media** – scene creation and VFX
- **E-commerce** – product display in 3D
- **Education** – interactive learning models
- **AR/VR** – immersive experiences

ADVANTAGES:

- Creates highly realistic 3D outputs
- Reduces manual modeling effort
- Easy to use with minimal technical skills
- Supports advanced visualization
- Enhances creativity and productivity

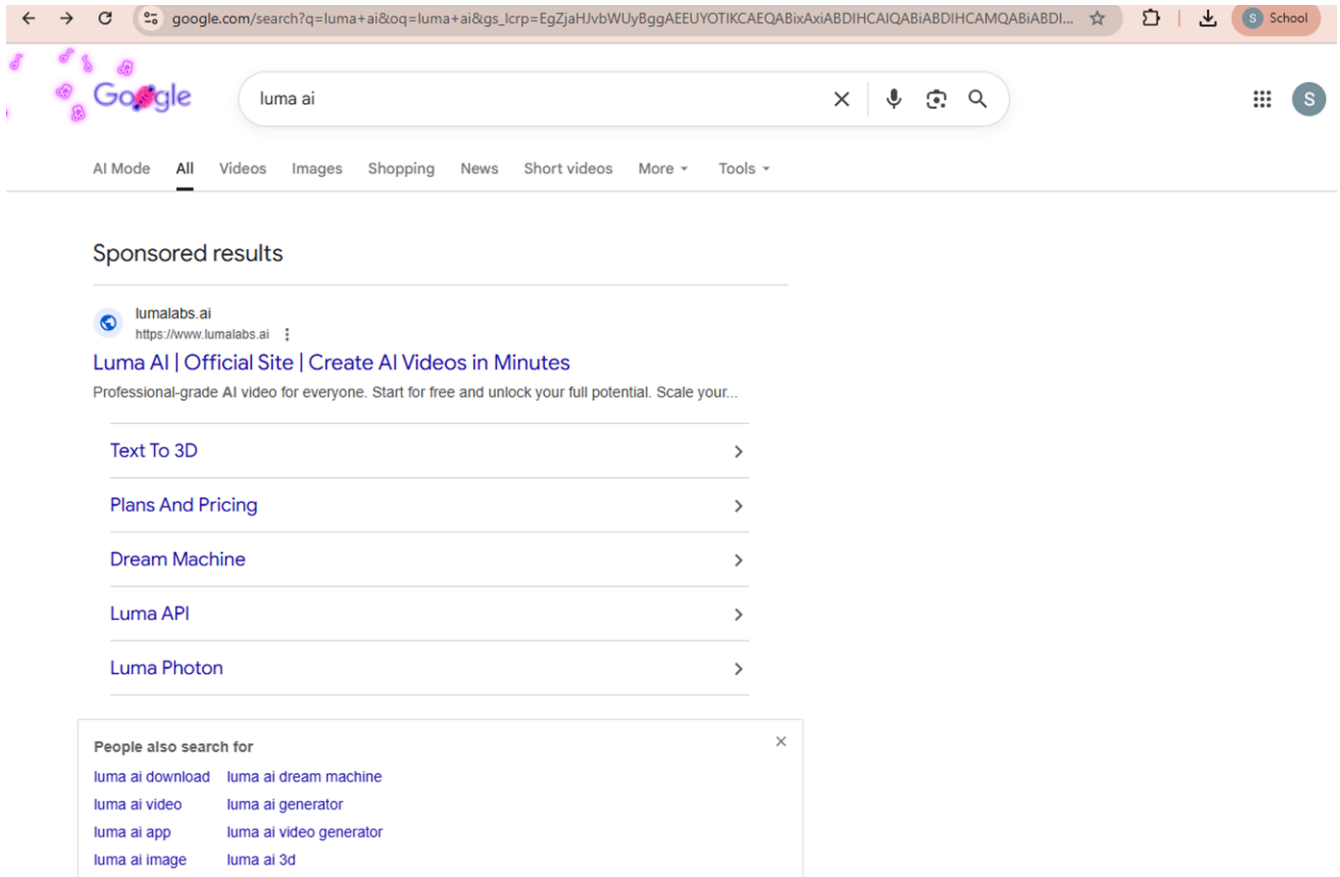
LIMITATIONS:

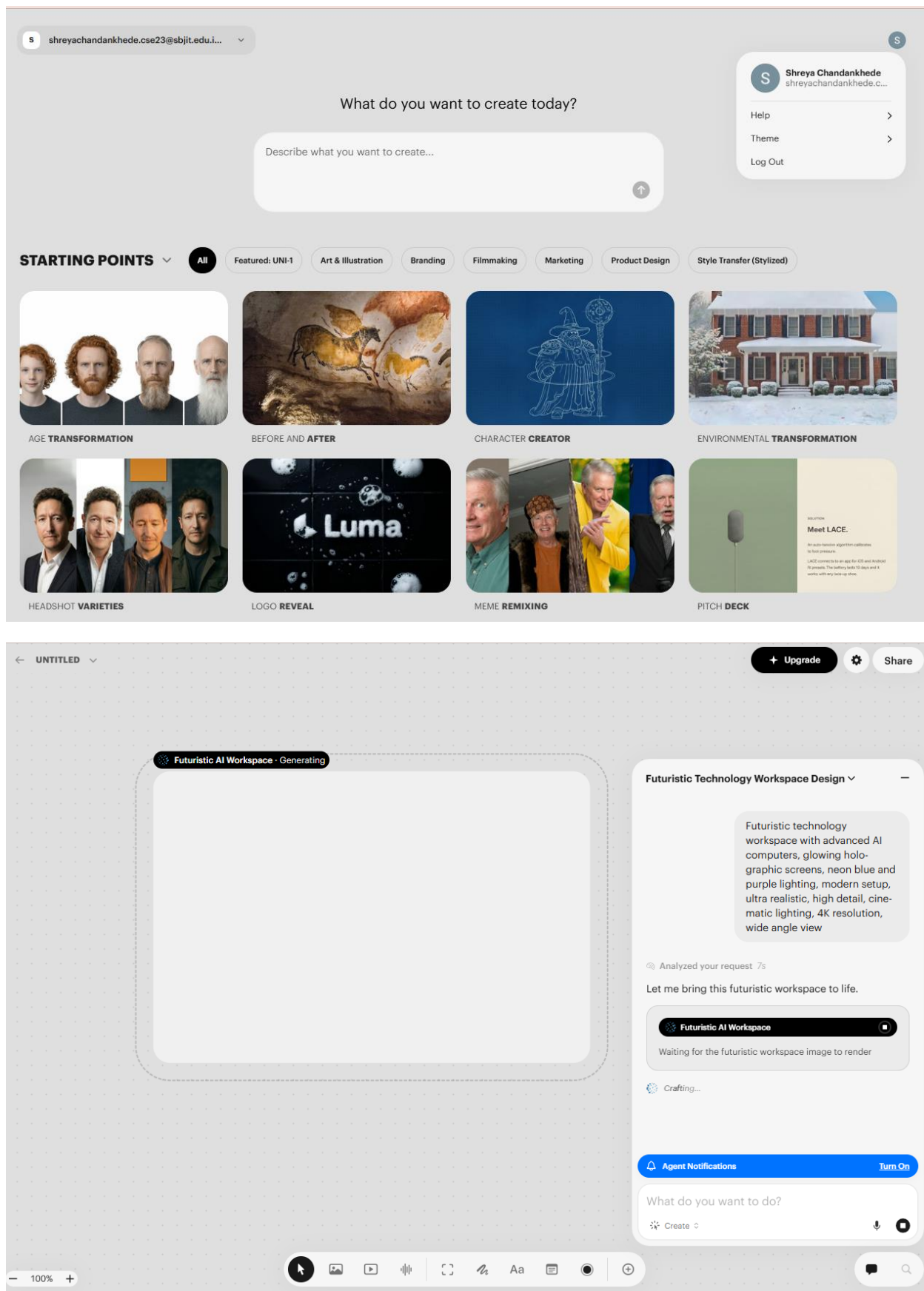
- Requires good input quality (images/videos)
- Processing may take time
- Limited free access/features
- Requires internet connection
- Hardware limitations for complex rendering

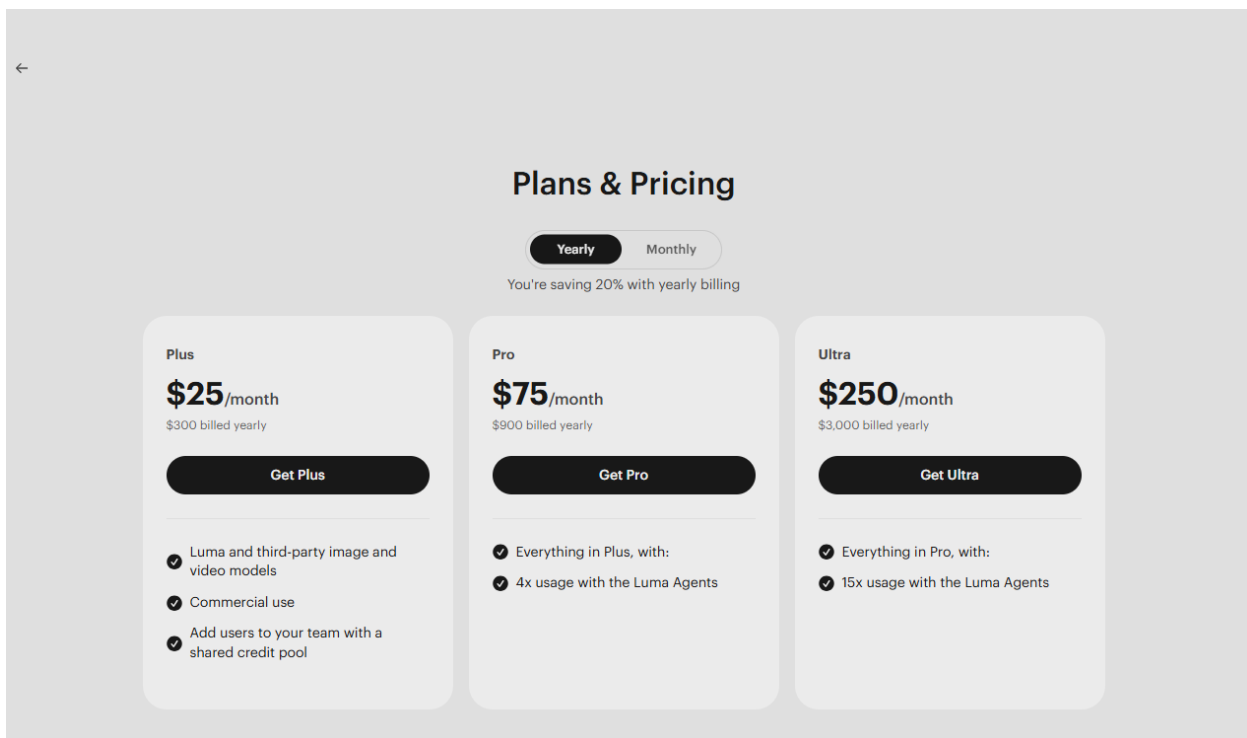
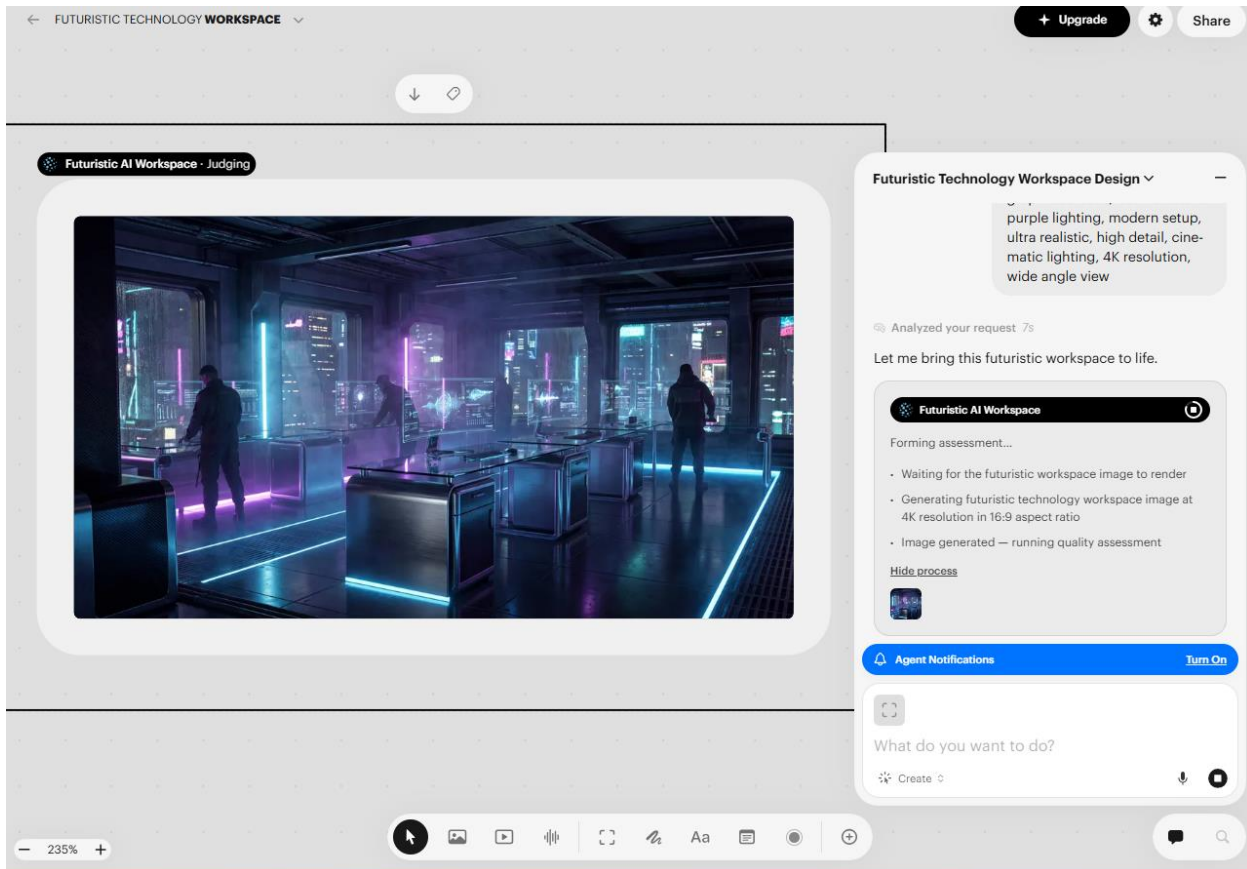
HOW TO USE LUMA AI (DETAILED STEPS):

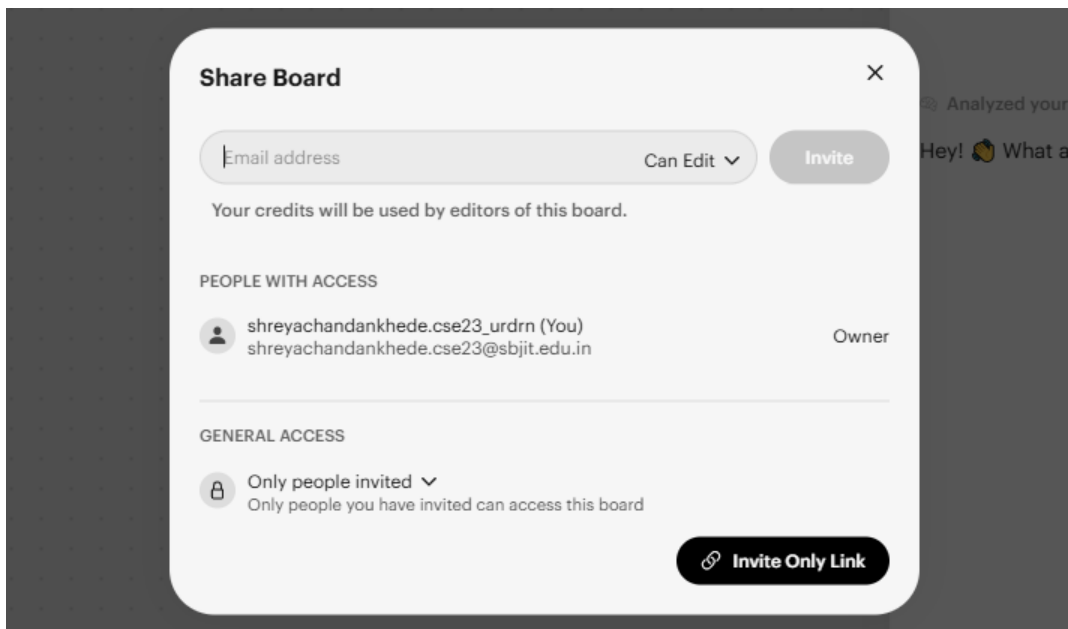
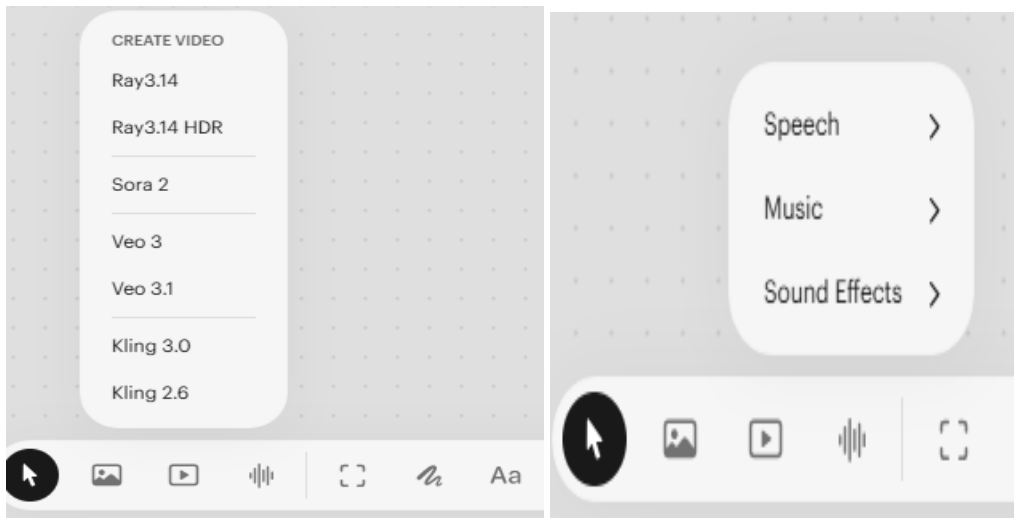
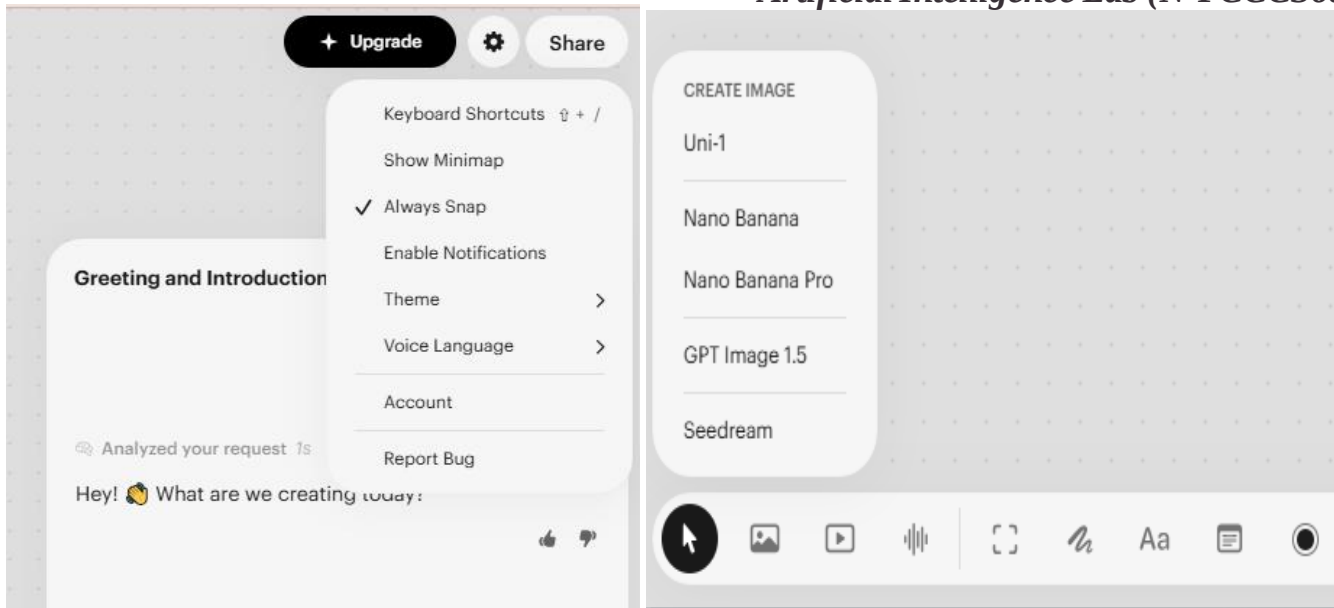
- **Step 1:** Open the official website or app
- **Step 2:** Sign up / Log in
- **Step 3:** Choose capture or generation option
- **Step 4:** Upload photos or record a video
- **Step 5:** Let AI process the scene
- **Step 6:** View generated 3D model
- **Step 7:** Edit or enhance output
- **Step 8:** Download or export the result
- **Step 9:** Improve by capturing better input or refining prompts

INPUT & OUTPUT:









CONCLUSION:

DISCUSSION QUESTIONS:

1. What is a prompt in AI image generation?

2. Why is prompt engineering important in tools like Luma AI?

3. How does Luma AI generate images or 3D models from input?

4. What are the real-world applications of AI-based image generation?

REFERENCES:

- <https://lumalabs.ai>
- <https://docs.lumalabs.ai>
- <https://lumalabs.ai/learn>